

HCSM-21

The Time Theorem

Complex Time from the Five-Layer Loop, the Topological Derivation of the Floor Function, and the Processor as Projection onto Nonzero Real Time

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Abstract

We derive time and its arrow in the Hodge Complex Standard Model (HCSM) as internal structures of the five layer loop. No external time coordinate is assumed. The loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$$

acts diagonally on the 14-mode multiplet. On each mode m it has the polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$. The complex time of mode m is

$$\tau_m \equiv \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m),$$

where ω_m is the mode's winding-generator frequency, $T_{\text{DME}} = 1/(2\gamma)$ is the DME relaxation time, and r_m is the mode's per pass amplitude. The real part θ_m is the clock; the imaginary part $-\ln r_m > 0$ is the arrow increment. The processor T is exactly the projection onto modes with $\text{Re}(\tau_m) > 0$. Time is simultaneously the measurement outcome and the selection criterion of the loop. Three algebraic bridges are established. (i) The floor function in the Core perturbation is the winding number of the DME-Core interface phase over one loop pass; it is not an ansatz. (ii) Dimensional homogeneity of τ_m is proved: both θ_m and $-\ln r_m$ are per-pass products of a rate with T_{DME} . (iii) The exact ratio $|M_A|/|M_B| = 3 = d - 1$ is a consequence of the closed-form T_{DME} from HCSM-54. We state the HCSM Time Theorem and identify what is derived, what is disclosed, and what is open. All numerical claims are verified in reproducible Python at machine precision.

Derivation status summary.

- **DERIVED** (substrate-level): the loop operator diagonalization; the complex time closed form $\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$; the closed forms of τ_A, τ_B, τ_C ; the multiplicative loop decay decomposition into Core and EFT factors; the topological derivation of the floor function from the DME-Core winding condition; dimensional homogeneity of τ_m ; the processor identity $T = \Pi_{\text{Re}(\tau)>0}$; the arrow of time from the Core Lyapunov functional; the exact ratio $|M_A|/|M_B| = 3$.
- **DISCLOSED INPUTS**: (A) the DME stability coefficient $\gamma = c_2/I(2)$ and time quantum $T_{\text{DME}} = 1/(2\gamma)$ from HCSM-54 [9]; (B) the EFT equation of state w and bulk viscosity $\zeta H/\rho$ from HCSM-56 [10].
- **OPEN**: the loop pass duration selection (T_{DME} vs. T_{beat}); the identification of continuous time $t = k T_{\text{DME}}$ from a deeper principle; the physical conversion of T_{DME} to seconds.

Contents

1	Introduction	3
1.1	The time problem	3
1.2	Summary of results	3
1.3	Derivation status	4
1.4	Organization	5
2	Setup: the five layer loop and its state spaces	5
2.1	State spaces	5
2.2	The loop operator	5
3	Level 1 — FDLM/WIN: winding phase	5
3.1	Substrate and modes	5
3.2	Winding generator	5
3.3	The frequency classes	6
3.4	What time is doing at Level 1	6
4	Level 2 — DME: time quantum and the topological floor derivation	6
4.1	The DME stability coefficient	6
4.2	The $1 \rightarrow 2$ transition: phase handoff	6
4.3	The $2 \rightarrow 3$ transition: cycle-quantized Core perturbation	6
4.4	What time is doing at Level 2	7
5	Level 3 — Core: the arrow of time	8
5.1	Flow equation	8
5.2	Lyapunov functional	8
5.3	Master conditions	8
5.4	The $3 \rightarrow 4$ transition	8
5.5	What time is doing at Level 3	8
6	Level 4 — EFT: damping	8
6.1	Mode equation	8
6.2	EFT parameters	9
6.3	Regime	9
6.4	What time is doing at Level 4	9
7	Level 5 — Processor T	9
7.1	Definition	9
7.2	Properties	9
7.3	Charge bookkeeping	9
7.4	What time is doing at Level 5	9
8	The complex time operator	10
8.1	Definition	10
8.2	Loop-decay decomposition	10
8.3	Dimensional homogeneity	10
8.4	Structural identities	11

9	The processor as projection onto nonzero real time	11
10	The exact ratio $M_A / M_B = 3$	11
11	The arrow of time	12
12	The HCSM Time Theorem	12
13	Status, disclosed inputs, and open items	12
13.1	Derived	12
13.2	Disclosed inputs	13
13.3	Open	13
14	Falsifiable predictions	13
15	Conclusion	14
A	Reproducible verification	15
B	Web Crawler Metadata	15
B.1	Dublin Core	15
B.2	Schema.org JSON-LD	16
B.3	BibTeX for Automated Indexing	17

1 Introduction

1.1 The time problem

The Standard Model has no fundamental arrow of time. Time is an external parameter, and its direction is accounted for by statistical mechanics. The Hodge Complex Standard Model (HCSM) supplies a closed five-layer loop with a preferred direction and a dissipative operator. In this paper we show that time is not an input to HCSM; it is a readout of the loop.

Two aspects of time emerge from the loop structure:

1. **Clock time** — the phase accumulated by M_{loop} per pass.
2. **Arrow time** — the directed decay of M_{loop}^k because $|M_j| < 1$.

The first gives clocks. The second gives the arrow. Both arise from one operator.

1.2 Summary of results

The present paper establishes the following.

1. (Theorem 8.2). On each observable mode m , the loop operator M_{loop} is diagonal with polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$, and the complex time is $\tau_m = \omega_m T_{\text{DME}} + i \ln r_m$ with closed form

$$\begin{aligned}\tau_A &= 2.0121430664919213 - 0.5521962501528914i, \\ \tau_B &= 6.0364291994757650 - 1.6508085388210012i, \quad \tau_C = 0.\end{aligned}$$

2. (Theorem 4.5). The floor function in the Core perturbation $\phi_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \cdot \delta_{\text{edge}}$ is the winding number of the DME–Core interface phase over one loop pass. It is not a postulated integer-floor rule.
3. (Proposition 8.3). The complex time τ_m is dimensionally homogeneous: both θ_m and $-\ln r_m$ are per pass quantities of the form $(\text{rate}) \times T_{\text{DME}}$, and both scale linearly with the Core flow parameter λ .
4. (Theorem 9.1). The processor $T = P_{\text{mesh}} + P_\tau$ is exactly the projection onto modes with nonzero real time:

$$T = \Pi_{\text{Re}(\tau) > 0}.$$

Kernel modes have $\tau_m = 0$ and are frozen under M_{loop} .

5. (Theorem 11.1). The arrow of time is the direction of increasing loop count k , equivalently the direction of decreasing Core Lyapunov functional $L[] = \gamma^2$, equivalently the direction of increasing $-\ln r_m$.

1.3 Derivation status

Derived (substrate-level).

- The loop operator diagonalization and the complex time formula (Theorem 8.2).
- The closed forms of τ_A, τ_B, τ_C .
- The multiplicative loop decay decomposition into Core and EFT factors.
- The topological derivation of the floor function from the DME-Core winding condition (Lemma 4.2, Theorem 4.5).
- Dimensional homogeneity of τ_m (Proposition 8.3).
- The processor identity $T = \Pi_{\text{Re}(\tau) > 0}$ (Theorem 9.1).
- The arrow of time from the Core Lyapunov functional (Theorem 11.1).
- The exact ratio $|M_A|/|M_B| = 3$ (restated from HCSM-20, Theorem 10.1).

Disclosed inputs.

- (A) The DME stability coefficient $\gamma = c_2/I(2)$ and time quantum $T_{\text{DME}} = 1/(2\gamma)$ from HCSM-54 [9].
- (B) The EFT equation of state $w = -1 - (1 - \gamma)^2/(2\pi)$ and bulk viscosity $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$ from HCSM-56 [10].

Open.

- The loop pass duration selection (T_{DME} vs. T_{beat}). Section 13.3 states this as an explicit open item.
- The identification of continuous time $t = k T_{\text{DME}}$ from a deeper principle.
- The physical conversion of T_{DME} to seconds.

1.4 Organization

Section 2 states the five-layer loop and its state spaces. Section 3 develops Level 1 (FDLM/WIN). Section 4 develops Level 2 (DME) and the topological floor derivation. Section 5 develops Level 3 (Core). Section 6 develops Level 4 (EFT). Section 7 develops Level 5 (Processor). Section 8 defines the complex-time operator and proves dimensional homogeneity. Section 9 proves the processor identity. Section 10 restates the exact ratio. Section 12 states the HCSM Time Theorem. Section 13 catalogues the derivation status. Section 14 lists falsifiable predictions. Appendix A provides reproducible Python verification.

2 Setup: the five layer loop and its state spaces

2.1 State spaces

Motion is the composition of five layers of the substrate, each with a well-defined state space and a unique operator (HCSM-20 [7]).

Layer	Name	State space	Operator type
1	FDLM/WIN	$\mathbb{C}^{14}$ mode vectors	Hermitian (unitary evolution)
2	DME	$\in \mathbb{R}$	scalar constraint
3	Core	(λ) trajectory	gradient flow
4	EFT	field configurations	damped wave equation
5	Processor T	$\mathbb{C}^{14}$ projected	orthogonal projection

2.2 The loop operator

The loop operator is

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (1)$$

On the mode basis, M_{loop} is diagonal:

$$M_{\text{loop}}(m) = r_m e^{i\theta_m}. \quad (2)$$

The five layers are not temporally sequential within a pass. They are algebraically composed. Time in the loop is read from the eigenvalues of the composition, not from the duration of individual layers.

3 Level 1 — FDLM/WIN: winding phase

3.1 Substrate and modes

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are listed in HCSM-04 [5] and HCSM-20 [7].

3.2 Winding generator

Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors. Define $M = n_1 n_2^T - n_2 n_1^T$, $\phi = (\pi/8)M$, and $H_{\text{wind}} = i \sin \phi$. The spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}, \quad (3)$$

with multiplicities 1, 1, 10, 1, 1 (HCSM-20, Corollary 3.5 [7]). The beat period is $T_{\text{beat}} = 2\pi/(2\sqrt{2})$. At $t = T_{\text{beat}}$, $e^{iH_{\text{wind}}T_{\text{beat}}} = I$.

3.3 The frequency classes

The 14 modes partition into three classes by frequency:

$$\omega_A = 2\sqrt{2} \text{ (3 modes)}, \quad \omega_B = 6\sqrt{2} \text{ (7 modes)}, \quad \omega = 0 \text{ (4 modes)}. \quad (4)$$

The ratio $\omega_B/\omega_A = 3$ is exact. The kernel class $\omega = 0$ is exactly the kernel of the processor T .

3.4 What time is doing at Level 1

Time at Level 1 is the phase of the winding generator. The eigenvalues of H_{wind} are the mode frequencies $\omega_m \in \{2\sqrt{2}, 6\sqrt{2}, 0\}$. The level is unitary; no amplitude is lost. It is a pure clock with an internal period T_{beat} .

4 Level 2 — DME: time quantum and the topological floor derivation

4.1 The DME stability coefficient

DME fixes the vacuum dimension $D = 4$ as a strict local maximum of the vacuum entropy (HCSM-54, Theorem 6.1 [9]). The stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188\dots, \quad (5)$$

with $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ and $c_2 = 2I''(2)$ given in closed form in HCSM-54. The relaxation time is

$$T_{\text{DME}} = \frac{1}{2\gamma} = 0.711400003516965862\dots \quad (6)$$

4.2 The $1 \rightarrow 2$ transition: phase handoff

The unique dimensionless quantity at the $1 \rightarrow 2$ boundary is

$$\Phi_{1 \rightarrow 2}(m) = \omega_m T_{\text{DME}}. \quad (7)$$

For class A , $\Phi_A = 2 + \delta_{\text{edge}}$; for class B , $\Phi_B = 6 + 3\delta_{\text{edge}}$, with

$$\delta_{\text{edge}} = \frac{\sqrt{2}}{\gamma} - 2 = 0.012143066491921761\dots \quad (8)$$

The residual δ_{edge} is the exact mismatch between the substrate and DME clocks.

4.3 The $2 \rightarrow 3$ transition: cycle-quantized Core perturbation

The transition rule ${}_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \cdot \delta_{\text{edge}}$ is often presented as a postulated integer-floor rule. We now derive it from a winding condition on the DME–Core interface.

Definition 4.1 (DME–Core interface). The DME–Core interface is a $U(1)$ -valued map from the loop parameter space S^1 (one loop pass) to the DME phase space,

$$\gamma_m : S^1 \rightarrow U(1), \quad \gamma_m(s) = e^{2\pi i \omega_m T_{\text{DME}} s}, \quad s \in [0, 1). \quad (9)$$

Lemma 4.2 (Lift and winding). *The lift of γ_m to $\mathbb{R}$ is the monotonically increasing function $\tilde{\gamma}_m(s) = \omega_m T_{\text{DME}} s$. Over one loop pass ($s : 0 \rightarrow 1$), the lift crosses integer values $k \in \mathbb{Z}$ exactly $\lfloor \omega_m T_{\text{DME}} \rfloor$ times (the final point excluded).*

Proof. The lift $\tilde{\gamma}_m$ ranges over $[0, \omega_m T_{\text{DME}})$. The integers in this half-open interval are $0, 1, \dots, \lfloor \omega_m T_{\text{DME}} \rfloor - 1$, counted with multiplicity 1 each. The remaining integer $\lfloor \omega_m T_{\text{DME}} \rfloor$ is the ceiling reached at the endpoint $s = 1$. Therefore the number of complete DME cycles per pass is $n_m \equiv \lfloor \omega_m T_{\text{DME}} \rfloor$. $\square$

Definition 4.3 (DME natural period). The DME layer has a natural phase period of 1 in the loop time unit T_{DME} . That is, one complete DME cycle corresponds to advancing the loop parameter by one unit of T_{DME} .

The natural period 1 in loop units is not imposed from outside; it is the definition of the DME relaxation cycle. Because the DME stability coefficient γ sets a single relaxation rate 2γ , one relaxation time $T_{\text{DME}} = 1/(2\gamma)$ is the natural DME cycle.

Definition 4.4 (DME–Core boundary functional). Define the DME–Core interface functional

$$J[0] = {}_0 + \Lambda \cdot \mathcal{W}[0 \leq |w + 1|], \quad (10)$$

with $\Lambda \rightarrow \infty$ enforcing the constraint ${}_0 > |w + 1|$.

Theorem 4.5 (Core perturbation quantization). *The unique minimizing ${}_0$ of J on the lattice $\delta_{\text{edge}} \mathbb{Z}_{>0}$, subject to the winding condition of Lemma 4.2, is*

$${}_0(m) = n_m \cdot \delta_{\text{edge}}, \quad n_m = \lfloor \omega_m T_{\text{DME}} \rfloor. \quad (11)$$

Proof. The functional J is finite only if ${}_0 > |w + 1|$. The lattice is $\delta_{\text{edge}} \mathbb{Z}_{>0}$. The winding condition fixes $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$. We verify that $n_m \cdot \delta_{\text{edge}} > |w + 1|$ for both non-trivial classes. Numerically, $|w + 1| = 0.014054078857664869$ and $\delta_{\text{edge}} = 0.012143066491921761$, so $|w + 1|/\delta_{\text{edge}} = 1.1573747757\dots$. Any $n \geq 2$ satisfies $n \cdot \delta_{\text{edge}} > |w + 1|$. For class A , $n_A = 2$, so ${}_0(A) = 2\delta_{\text{edge}} = 0.0242861\dots > |w + 1|$. For class B , $n_B = 6$, so ${}_0(B) = 6\delta_{\text{edge}} = 0.0728584\dots > |w + 1|$. Thus the winding-quantized ${}_0$ automatically satisfies the boundary constraint for both classes, and the minimizing element on the lattice is unique. $\square$

Remark 4.6 (The floor function is not an ansatz). The floor function is the winding number of the DME phase over one loop pass, computed on the $U(1)$ -valued interface. The boundary constraint ${}_0 > |w + 1|$ is a consistency condition automatically satisfied by the winding-quantized value.

4.4 What time is doing at Level 2

Level 2 has no internal dynamics. It imposes the time quantum T_{DME} at the $1 \rightarrow 2$ boundary and quantizes the Core perturbation by the DME phase winding number. This is the only level where the time unit crosses a transition. The output to Core is

$$\{T_{\text{DME}}, n_m = \lfloor \omega_m T_{\text{DME}} \rfloor, {}_0(m) = n_m \delta_{\text{edge}}, S''(0)/S_0 = -2\gamma\}. \quad (12)$$

5 Level 3 — Core: the arrow of time

5.1 Flow equation

Core obeys

$$\frac{d}{d\lambda} = -2\gamma. \quad (13)$$

The solution is $(\lambda) =_0 e^{-2\gamma\lambda}$.

5.2 Lyapunov functional

$$L[] = \gamma^2, \quad \frac{dL}{d\lambda} = -4\gamma^2 \leq 0, \quad (14)$$

strict for $\neq 0$. The Core flow is strictly monotone.

5.3 Master conditions

- M1 (monotonicity): $dL/d\lambda \leq 0$. Satisfied.
- M2 (irreversibility): under $\lambda \rightarrow -\lambda$, L increases. Satisfied.
- No oscillation: characteristic equation $s + 2\gamma = 0$ has real root $s = -2\gamma$.

5.4 The $3 \rightarrow 4$ transition

The Core-to-EFT transition occurs when $(\lambda_{\text{EFT}}) = |w + 1|$:

$$\lambda_{\text{EFT}}(m) = \frac{\ln(0(m)/|w + 1|)}{2\gamma}. \quad (15)$$

For class A : $\lambda_{\text{EFT}} = 0.389130725768564\dots$ For class B : $\lambda_{\text{EFT}} = 1.170683511790840\dots$

5.5 What time is doing at Level 3

Time at Level 3 is the flow parameter λ . The state decays monotonically. The Lyapunov functional decreases. This is the source of the arrow: the direction of increasing λ is the direction of decreasing L . No statistical assumption is used.

6 Level 4 — EFT: damping

6.1 Mode equation

The EFT mode equation is

$$\ddot{\phi} + \Gamma_{\text{eff}}\dot{\phi} + \left(\frac{k^2}{a^2} + m_{\text{eff}}^2\right)\phi = 0, \quad (16)$$

with $\Gamma_{\text{eff}} = (3 + \zeta H/\rho)H_{\text{local}}$ and $m_{\text{eff}}^2 = m_s^2 + \zeta R$.

6.2 EFT parameters

From HCSM-56 [10]:

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi}, \quad \frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi}. \quad (17)$$

The EFT decay factor per beat period is

$$D_{\text{factor}} = \exp\left(-\frac{\zeta H T_{\text{beat}}}{\rho}\right) = 0.994810128563526019 \dots \quad (18)$$

6.3 Regime

$$\frac{\zeta R}{m_s^2} = 5.9 \times 10^{-26}, \quad \frac{\Gamma_{\text{eff}}}{\omega_0} = 5.2 \times 10^{-13}, \quad \frac{k}{m_s} < 2.7 \times 10^{-13}. \quad (19)$$

The EFT is extremely underdamped, mass-dominated, gradient-negligible. All 14 modes oscillate at the same frequency m_s and decay at the same rate $\Gamma_{\text{eff}}/2$.

6.4 What time is doing at Level 4

Time at Level 4 is the damped-oscillation time. The field oscillates at ω_m and damps at $\Gamma_{\text{eff}}/2$. The EFT is a coarse-grained description; it does not resolve individual loop passes.

7 Level 5 — Processor T

7.1 Definition

$$T = P_{\text{mesh}} + P_{\tau}, \quad (20)$$

where P_{mesh} projects onto the nine mesh-resonance modes $\{\nu, e, u, d, s, \mu, c, b, t\}$ and P_{τ} projects onto the rank-1 mode τ (HCSM-20, Section 8 [7]).

7.2 Properties

$T^2 = T$, $T^\dagger = T$, $\text{rank } T = 10$. Kernel: $\{\text{dark}, W, Z, H\}$.

7.3 Charge bookkeeping

$$\sum_{j \in \text{image}(T)} Q_j = -2, \quad \sum_{j \in \text{ker}(T)} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1. \quad (21)$$

7.4 What time is doing at Level 5

Time at Level 5 is the selection criterion. The processor reads the complex time of each mode and selects those with $\text{Re}(\tau_m) > 0$. Modes with $\tau_m = 0$ are killed. The kernel modes are frozen under M_{loop} .

8 The complex time operator

8.1 Definition

Definition 8.1 (Complex time). For each mode m in the observable sector, the complex time is

$$\tau_m \equiv \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m). \quad (22)$$

The real part $\text{Re}(\tau_m) = \omega_m T_{\text{DME}}$ is the clock phase per pass; the imaginary part $-\text{Im}(\tau_m) = -\ln r_m > 0$ is the arrow increment per pass.

Theorem 8.2 (Closed form of τ_m). *For the two non-trivial classes,*

$$\tau_A = 2.0121430664919213 - 0.5521962501528914 i, \quad (23)$$

$$\tau_B = 6.0364291994757650 - 1.6508085388210012 i. \quad (24)$$

The kernel has $\tau_C = 0$.

Proof. Direct from $\omega_A = 2\sqrt{2}$, $\omega_B = 6\sqrt{2}$, $T_{\text{DME}} = 1/(2\gamma)$, and r_A, r_B computed from the loop-decay decomposition below. $\square$

8.2 Loop-decay decomposition

The loop amplitude decomposes exactly as

$$r_m = \underbrace{\frac{|w+1|}{0(m)}}_{\text{Core}} \cdot \underbrace{D_{\text{factor}}}_{\text{EFT}}. \quad (25)$$

Numerically,

$$r_A = 0.5786873879 \times 0.9948101286 = 0.575684074716075633, \quad (26)$$

$$r_B = 0.1928957960 \times 0.9948101286 = 0.191894691572025183. \quad (27)$$

The Core and EFT contributions compose multiplicatively.

8.3 Dimensional homogeneity

Proposition 8.3 (Dimensional homogeneity). *Both θ_m and $-\ln r_m$ are per-pass quantities of the form (rate) $\times T_{\text{DME}}$. Under the Core flow parameter λ , both quantities scale linearly with λ .*

Proof. The phase accumulated over one loop pass is $\theta_m = \omega_m T_{\text{DME}}$, which is the product of the mode frequency ω_m (units: inverse time) and the loop time unit T_{DME} (units: time).

The log-amplitude decrement per pass is $-\ln r_m$. Define $\Gamma_m \equiv -2 \ln r_m / T_{\text{DME}}$. Then $-\ln r_m = (\Gamma_m/2) T_{\text{DME}}$, which is the product of the decay rate $\Gamma_m/2$ (units: inverse time) and T_{DME} .

Both θ_m and $-\ln r_m$ therefore have the form (rate) $\times T_{\text{DME}}$. Both are dimensionless. Both are linear in the continuous time coordinate $t = k T_{\text{DME}}$, since after k passes $\theta_m \cdot k = \omega_m t$ and $-\ln r_m \cdot k = (\Gamma_m/2) t$.

Under the Core flow parameter λ , the Core amplitude decays as $(\lambda) =_0 e^{-2\gamma\lambda}$, so $\ln(\lambda) = \ln_0 - 2\gamma\lambda$ is linear in λ . The per-pass log decrement $-\ln r_m$ is therefore proportional to the Core flow parameter, and the phase θ_m is likewise linear in the loop parameter. The two quantities share the same scaling dimension and the same affine dependence on the flow parameter.

Therefore the additive combination $\tau_m = \theta_m + i \ln r_m$ is dimensionally homogeneous. $\square$

8.4 Structural identities

$$\frac{\theta_B}{\theta_A} = 3 \text{ (exact)}, \quad \ln \frac{r_A}{r_B} = \ln 3 \text{ (exact)}. \quad (28)$$

9 The processor as projection onto nonzero real time

Theorem 9.1 (Processor = projection onto nonzero real time). *Let $T = P_{\text{mesh}} + P_\tau$ be the processor. Then*

$$T = \Pi_{\text{Re}(\tau) > 0}, \quad (29)$$

i.e. T keeps exactly those modes with $\text{Re}(\tau_m) > 0$ and kills those with $\tau_m = 0$.

Proof. By direct computation on the 14-mode multiplet: every mode in the image of T (that is, every mode with $\delta \neq 0$ or $s = +1$) has $\omega_m \in \{2\sqrt{2}, 6\sqrt{2}\}$, hence $\text{Re}(\tau_m) = \omega_m T_{\text{DME}} > 0$. Every mode in the kernel of T (that is, every mode with $\delta = 0$ and $s \neq +1$) has $\omega_m = 0$, hence $\tau_m = 0$. $\square$

Remark 9.2. The image of T is exactly the set of modes for which the loop produces nonzero complex time. The kernel of T is exactly the set of modes for which the loop produces no time.

10 The exact ratio $|M_A|/|M_B| = 3$

Theorem 10.1 (Exact ratio). *The ratio of the two non-trivial loop-eigenvalue magnitudes is exactly 3:*

$$\frac{|M_A|}{|M_B|} = \frac{r_A}{r_B} = 3 = d - 1. \quad (30)$$

The framework constants $|w + 1|$, δ_{edge} , and D_{factor} do not appear in the ratio.

Proof (restated from HCSM-20, Theorem 7.1). From the loop-decay decomposition,

$$r_m = \frac{|w + 1|}{o(m)} D_{\text{factor}}, \quad o(m) = n_m \delta_{\text{edge}}, \quad n_m = \lfloor \omega_m T_{\text{DME}} \rfloor. \quad (31)$$

Therefore

$$\frac{r_A}{r_B} = \frac{n_B}{n_A}. \quad (32)$$

With $\gamma = c_2/I(2)$ from HCSM-54,

$$\omega_A T_{\text{DME}} = \frac{\sqrt{2}}{\gamma} = 2 + \delta_{\text{edge}}, \quad \omega_B T_{\text{DME}} = \frac{3\sqrt{2}}{\gamma} = 6 + 3\delta_{\text{edge}}, \quad (33)$$

and $0 < \delta_{\text{edge}} < 1/3$ gives $n_A = 2$, $n_B = 6$. Hence $r_A/r_B = 6/2 = 3$. $\square$

Corollary 10.2. *Since $|M_{\text{loop}}(m)| = r_m$ for observable m , we have $|M_A|/|M_B| = 3$.*

11 The arrow of time

Theorem 11.1 (Arrow of time). *The arrow of time is the direction of increasing loop count k , equivalently the direction of decreasing Core Lyapunov functional $L[] = \gamma^2$, equivalently the direction of increasing $-\ln r_m$.*

Proof. At Level 1, the FDLM layer is unitary; no amplitude is lost. At Level 3, the Core flow $d/d\lambda = -2\gamma$ is strictly monotone (equation (13)) and the Lyapunov functional (14) decreases strictly for $\neq 0$. At Level 4, the EFT contributes the per-beat decay factor $D_{\text{factor}} < 1$. Hence for each surviving mode m , $r_m < 1$ and $-\ln r_m > 0$. Increasing k increases $-\ln r_m$, decreases r_m , and decreases the Lyapunov functional. The three directions agree. $\square$

12 The HCSM Time Theorem

Theorem 12.1 (HCSM Time Theorem). *In the five-layer HCSM:*

1. *No external time coordinate is assumed. The loop operator M_{loop} is the only time-evolution object.*
2. *On each observable mode m , $M_{\text{loop}}(m) = r_m e^{i\theta_m}$ with $\theta_m = \omega_m T_{\text{DME}}$ and $T_{\text{DME}} = 1/(2\gamma)$.*
3. *The complex time per pass is $\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$. Both components share the scaling dimension (rate) $\times T_{\text{DME}}$ (Proposition 8.3).*
4. *Discrete HCSM time is the loop count $k \in \mathbb{Z}_{\geq 0}$. Continuous HCSM time is $t = kT_{\text{DME}}$. Under this identification, the loop evolution is*

$$|\psi_m(t)\rangle = e^{i\omega_m t} e^{-\Gamma_m t/2} |e_m\rangle, \quad \Gamma_m = \frac{-2 \ln r_m}{T_{\text{DME}}} > 0. \quad (34)$$

5. *The arrow of time is the direction of increasing k , equivalently the direction of decreasing Core Lyapunov functional $L[] = \gamma^2$, equivalently the direction of increasing $-\ln r_m$ (Theorem 11.1).*
6. *The processor T is exactly the projection onto modes with nonzero real time: $T = \Pi_{\text{Re}(\tau)>0}$ (Theorem 9.1).*
7. *Kernel modes have $\tau_m = 0$ and are frozen under M_{loop} .*
8. *Exact ratio: $|M_A|/|M_B| = 3$ (Theorem 10.1).*
9. *The floor function in the Core perturbation $_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \delta_{\text{edge}}$ is the winding number of the DME–Core interface phase over one loop pass (Lemma 4.2, Theorem 4.5).*

Proof. Parts (1)–(9) are established in Theorems 8.2, 4.5, 8.3, 9.1, 10.1, and 11.1 above. $\square$

13 Status, disclosed inputs, and open items

13.1 Derived

1. Loop operator diagonalization and complex-time formula (Definition 8.1, Theorem 8.2).
2. Closed forms of τ_A, τ_B, τ_C (Theorem 8.2).
3. Multiplicative decomposition of the loop decay into Core and EFT factors (equation (25)).

4. Topological derivation of the floor function from the DME–Core winding condition (Lemma 4.2, Theorem 4.5).
5. Dimensional homogeneity of τ_m (Proposition 8.3).
6. Processor identity $T = \Pi_{\text{Re}(\tau) > 0}$ (Theorem 9.1).
7. Arrow of time from Core Lyapunov and loop dissipativity (Theorem 11.1).
8. Exact algebraic ratio $|M_A|/|M_B| = 3$ (Theorem 10.1).
9. Composition order forced by state-space chain (Section 2).

13.2 Disclosed inputs

1. The DME stability coefficient $\gamma = c_2/I(2)$ and time quantum $T_{\text{DME}} = 1/(2\gamma)$ from HCSM-54 [9]. In HCSM-54, γ is derived from the AGM closed form $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ and the closed form of $c_2 = 2I''(2)$.
2. The EFT equation of state $w = -1 - (1 - \gamma)^2/(2\pi)$ and bulk viscosity $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$ from HCSM-56 [10].

13.3 Open

1. **The loop pass duration selection.** Both T_{DME} (DME relaxation) and T_{beat} (Level-1 internal period) are dimensionless time-like quantities. The transition rule uses T_{DME} ; Level 1 uses T_{beat} internally. The framework identifies the loop pass duration with T_{DME} by the transition rule, but does not derive from a deeper principle that T_{DME} and not T_{beat} is the physical pass duration. The ratio $T_{\text{beat}}/T_{\text{DME}} = 3.1226\dots$ is a structural constant. This is the same open item as in HCSM-20.
2. **The identification of continuous time.** The relation $t = kT_{\text{DME}}$ is stated as a correspondence between the discrete loop count and a continuous parameter. A derivation from a deeper principle is open.
3. **The physical conversion.** The framework measures time in units of T_{DME} . Conversion to seconds requires choosing a physical process to calibrate against. This choice is external to the framework.

Remark 13.1 (Lorentzian signature is now closed). In the prior WIN-era Time Theorem, the Lorentzian signature $(-, +, +, +)$ was listed as an open item. In HCSM, it has been derived from the Hodge star on $\ker D$ (HCSM-03 [4]), with the timelike direction $(v_0 + v_2)/\sqrt{2}$. This item is therefore closed at the HCSM level.

14 Falsifiable predictions

1. **Complex time.** $\tau_A = 2.0121430664919213 - 0.5521962501528914i$ and $\tau_B = 6.0364291994757650 - 1.6508085388210012i$, exactly. If a future computation or measurement of the framework's complex time for either class gives a value outside 10^{-10} of these, the Time Theorem is falsified.
2. **Processor identity.** $T = \Pi_{\text{Re}(\tau) > 0}$. If a future mode in the framework's 14-mode multiplet is found with $\text{Re}(\tau) > 0$ but outside the image of T , or vice versa, the identity is falsified.

3. **Arrow direction.** The arrow points in the direction of increasing $-\ln r_m$. A detection of a loop mode with $|M_j| > 1$ would falsify the dissipativity theorem.
4. **Kernel freezing.** Kernel modes $\{\text{dark}, W, Z, H\}$ have $\tau_m = 0$ and are exactly frozen under M_{loop} . Any detection of nontrivial time evolution of these modes would falsify the Time Theorem.

15 Conclusion

Time in the Hodge Complex Standard Model is not an external coordinate. It is the complex eigenvalue of the loop operator,

$$\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m). \quad (35)$$

Its real part is the clock. Its imaginary part is the arrow. The processor T is the projection onto nonzero real time. The loop observes itself by reading its own complex time. No external time unit is required: one pass through the loop advances HCSM time by one unit of T_{DME} .

Three algebraic bridges are closed in this paper. First, the floor function of the Core perturbation is the winding number of the DME–Core interface (Lemma 4.2, Theorem 4.5); it is not an ansatz. Second, the exact ratio $|M_A|/|M_B| = 3$ is exact by algebraic cancellation of framework constants (Theorem 10.1), with the 3 matching the Standard Model generation count $d - 1$. Third, dimensional homogeneity of the complex time follows because both θ_m and $-\ln r_m$ are per-pass products of a rate with T_{DME} (Proposition 8.3).

The present paper follows the HCSM-06/07 pattern. It derives everything in its own domain that can be derived, discloses the two physical inputs (γ and $w, \zeta H/\rho$) explicitly, and names the specific open items (loop pass duration, continuous-time identification, physical conversion) that future papers must close. The framework’s strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

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A Reproducible verification

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-21_verification`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the 14-mode multiplet, the winding generator, and the loop operator, and verifies:

1. The loop operator diagonalization (Theorem 8.2).
2. The closed forms of τ_A, τ_B, τ_C (Theorem 8.2).
3. The multiplicative loop-decay decomposition (equation (25)).
4. The topological derivation of the floor function (Lemma 4.2, Theorem 4.5).
5. Dimensional homogeneity of τ_m (Proposition 8.3).
6. The processor identity $T = \Pi_{\text{Re}(\tau) > 0}$ (Theorem 9.1).
7. The arrow of time from Core Lyapunov (Theorem 11.1).
8. The exact ratio $|M_A|/|M_B| = 3$ (Theorem 10.1).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

B Web Crawler Metadata

B.1 Dublin Core

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DME-Core winding condition; processor projection; D4 symmetry
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function of the Core perturbation is the winding number of the
DME-Core interface over one loop pass, not an ansatz. Dimensional
homogeneity of τ_m is proved.
DC.identifier: HCSM-21-2026-09

B.2 Schema.org JSON-LD

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B.3 BibTeX for Automated Indexing

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